

# JUVENTUS FC



## Project Report

**Course:** CSCI390: Web Programming

**Academic Year:** Spring 2025-2026

**Submission Date:** May 31, 2026

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**GitHub Repository:** <https://github.com/hussein112001/juventus-fc-website-phase2-final>

**Live Website:** <https://hussein112001.github.io/juventus-fc-website-phase2-final/>

## 1. Executive Summary

This report documents the development of the Juventus FC web portal for Phase 2 of the CSCI390 course. The project involves a complete transition from static HTML to a modern, dynamic **ReactJS** application. The focus was on creating a professional, responsive, and high-performance frontend using the latest industry standards.

## 2. System Design

The application follows a modular, component-based architecture to ensure scalability and ease of maintenance. Key architectural decisions include:

- **Component-Based UI:** Reusable elements like Navigation, Footer, and Player Cards.
- **Client-Side Routing:** Seamless page transitions without browser reloads using Wouter.
- **Type-Safe Development:** Using TypeScript to prevent runtime errors and improve code quality.
- **Responsive Grid:** A mobile-first approach using Tailwind CSS for all device sizes.

### 3. Technologies Used

| Category           | Technology     | Purpose                                           |
|--------------------|----------------|---------------------------------------------------|
| Frontend Framework | React 19       | Core library for building the user interface.     |
| Language           | TypeScript     | Ensures type safety and robust code structure.    |
| Styling            | Tailwind CSS 4 | Modern utility-first CSS for professional design. |
| Icons              | Lucide React   | High-quality iconography for better UX.           |
| Build Tool         | Vite           | Ultra-fast bundling and development server.       |
| Routing            | Wouter         | Minimalist routing for React applications.        |

### 4. Key Implementation Features

- **Branding:** Custom Juventus theme with Black, White, and Gold color palette.
- **Squad Management:** Dynamic rendering of player profiles with detailed statistics.
- **Interactive Forms:** Functional contact form with real-time validation and toast notifications.
- **Responsive Layout:** Optimized views for Mobile, Tablet, and Desktop devices.

## 5. Code Snippets

### 5.1 Application Routing (App.tsx)

```
export default function App() {
  return (
    <div className="min-h-screen bg-background">
      <Navigation />
      <main>
        <Switch>
          <Route path="/" component={Home} />
          <Route path="/about" component={About} />
          <Route path="/squad" component={Squad} />
          <Route path="/contact" component={Contact} />
          <Route component={NotFound} />
        </Switch>
      </main>
      <Footer />
    </div>
  );
}
```

### 5.2 Navigation Logic

```
const [location] = useLocation();
const isActive = (path: string) => location === path;

return (
  <nav className="flex gap-4">
    <Link href="/">
      <a className={isActive("/") ? "active" : ""}>Home</a>
    </Link>
    {/* Other links */}
  </nav>
)
```

```
</nav>  
) ;
```

## 6. Challenges & Solutions

One of the primary challenges was migrating the static assets from Phase 1 into a React-friendly structure while maintaining the brand's visual integrity. This was solved by implementing a robust asset management system and using CSS variables for consistent branding.

## 7. Conclusion

The Phase 2 development of the Juventus FC portal successfully meets all academic and technical requirements. The application demonstrates a professional mastery of ReactJS, providing a solid foundation for future enhancements like backend integration and real-time updates.

**End of Report**